

Rockborne

Lasso Regression

Shrinking coefficients to zero for automatic feature selection

Objectives



What is Lasso

The L1 penalty and feature selection.



Lasso vs Ridge

Zeroing vs shrinking coefficients.



Choosing alpha

Balancing sparsity and accuracy.

Introduction



Intro to Lasso Regression

What is Lasso Regression?

What is it?

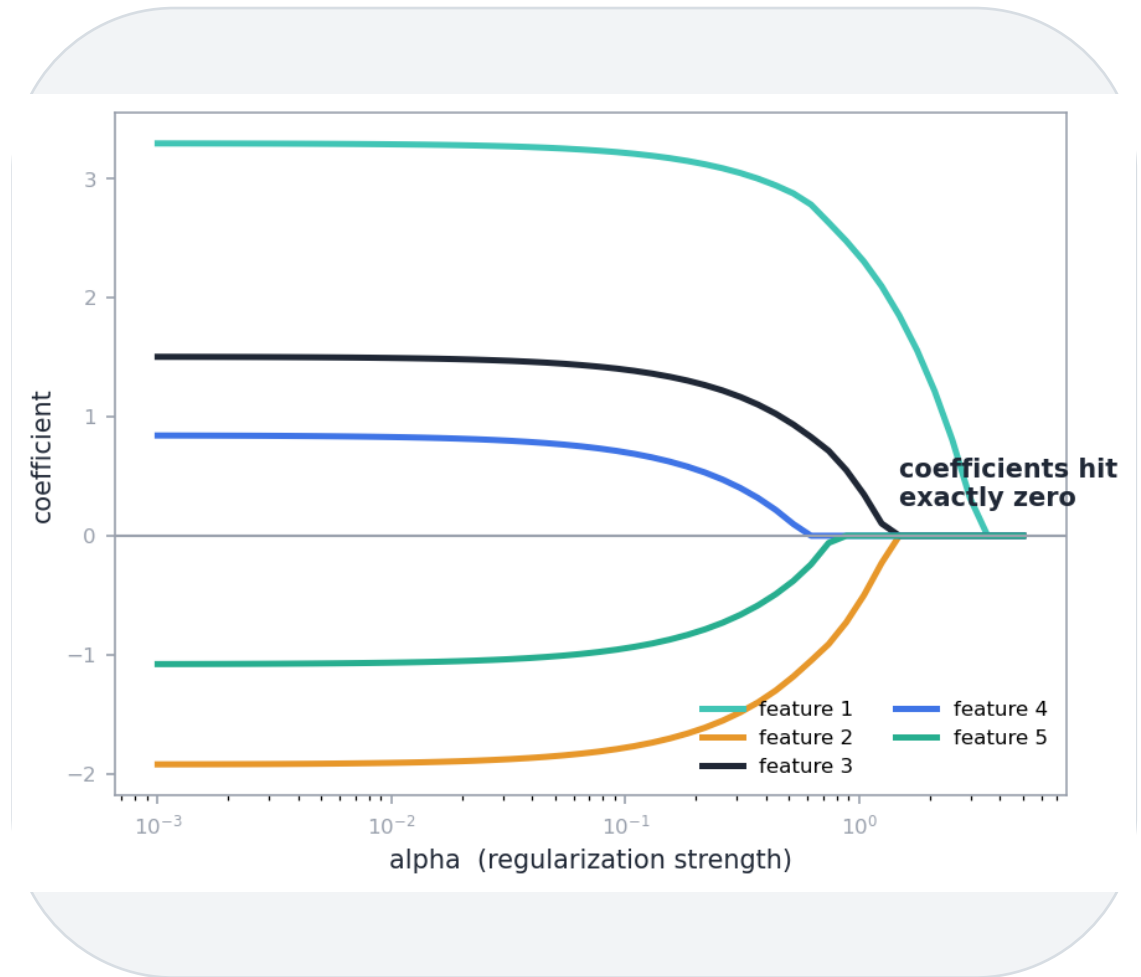
- Linear regression with an L1 penalty.
- The penalty is the sum of the absolute coefficients.
- Unlike Ridge, it can set coefficients to exactly zero.

When do we use it?

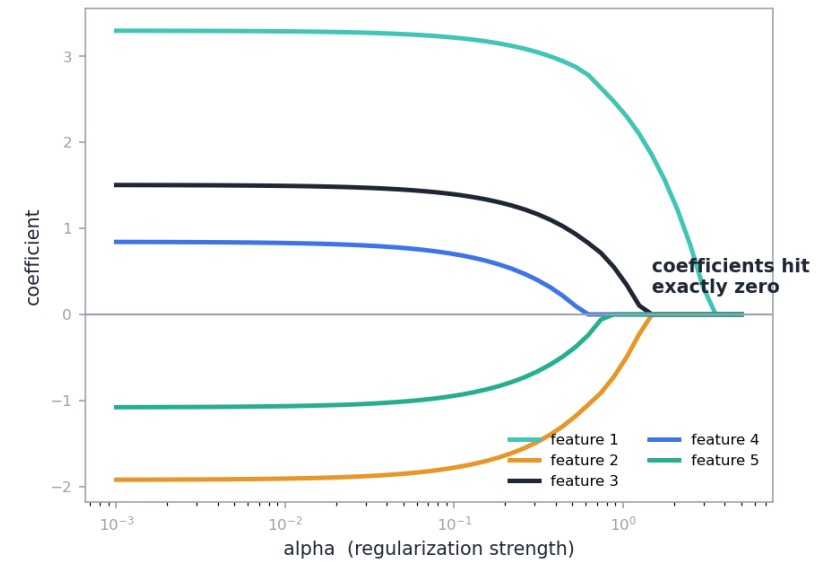
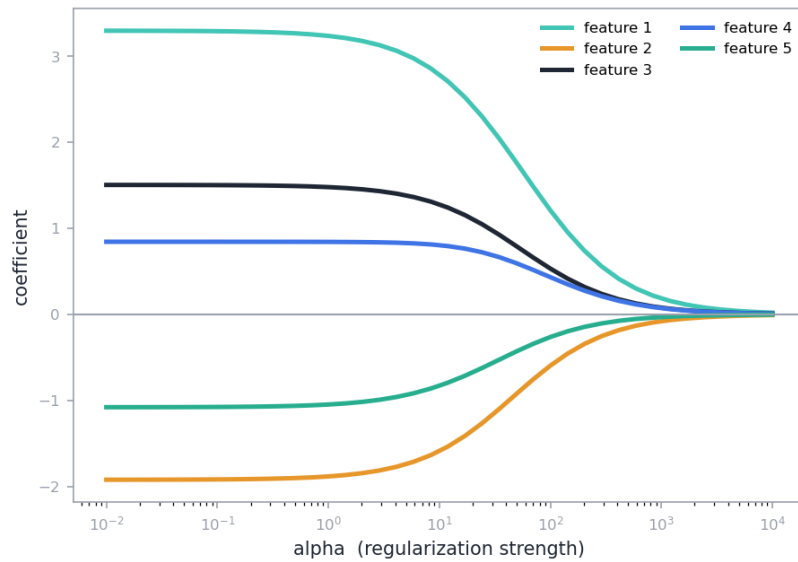
- When you have many features and want a simpler model.
- For automatic feature selection.
- When you want a sparse, interpretable set of drivers.

Automatic Feature Selection

- The L1 penalty drives some coefficients to exactly zero.
- Those features drop out of the model automatically.
- The result is a simpler, more interpretable model.
- Fewer, clearer drivers, ideal for a scorecard.



Lasso vs Ridge



Ridge (L2) shrinks all coefficients toward zero...

...but never exactly to zero (keeps every feature).

Lasso (L1) sets some coefficients to exactly zero.

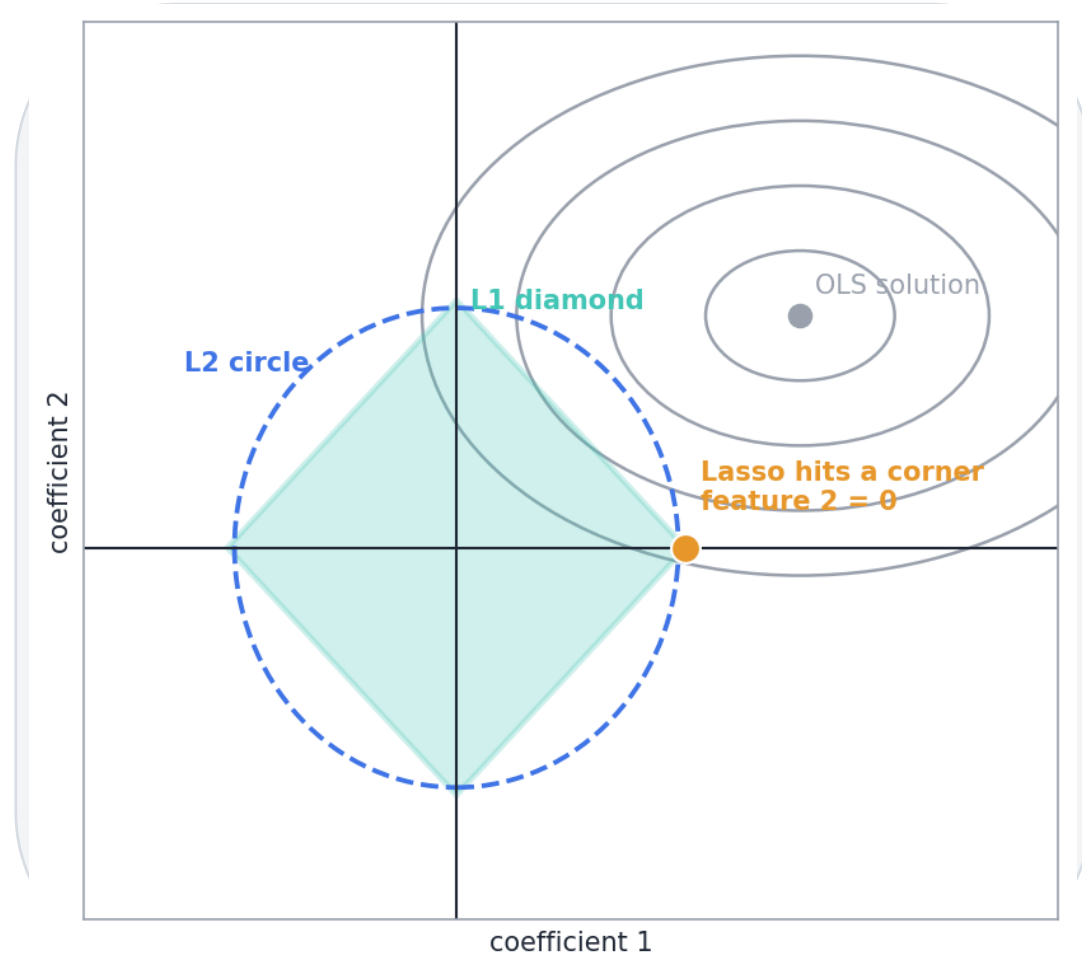
Use Lasso for a sparse model; Ridge to keep all features.

Lasso Regression



The Equation and Geometry

The Lasso Equation



1

OLS minimises RSS (the sum of squared residuals).

2

Lasso minimises $\text{RSS} + \alpha \times (\text{sum of } |\text{coefficients}|)$, the L1 penalty.

3

There is no closed form: it is solved by coordinate descent (hence `max_iter`).

4

The diamond's corners lie on the axes, so the solution often sets a coefficient to exactly zero.

Choosing Alpha



alpha = strength

A larger alpha zeros more coefficients.



LassoCV

Picks alpha by cross-validation.



Scale first

So the penalty is fair; too large an alpha under-fits.

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Time to get hands-on in Python

We move from theory to practice with a guided walkthrough you run yourself in Python.



Lasso Regression - Guided Walkthrough



Then: Regression Capstone

In this lesson, we explored



What is Lasso

Linear regression with an L1 penalty.



Lasso vs Ridge

Lasso zeros coefficients; Ridge only shrinks them.



The Equation & alpha

$RSS + L1 \text{ penalty}$; tune alpha with cross-validation.

Digging Deeper?



[Lasso \(API\)](#)

scikit-learn



[Lasso \(User Guide\)](#)

scikit-learn



[Lasso Regression in Python](#)

Statology

Questions?